

Q3		
(a)	Let mass of Planet Z be M, mass of argon molecules be m, Consider an argon molecule escapes from the planet's surface to infinity. By conservation of energy, Loss in KE = Gain in GPE $KE_{surface} - KE_{\infty} = U_{\infty} - U_{surface}$ $\frac{1}{2}mv^2 - 0 = 0 - \left(-\frac{GMm}{r} \right)$ $v = \sqrt{\frac{2GM}{r}}$ $= \sqrt{\frac{2G\left(\frac{4}{3}\pi r^3\rho\right)}{r}}$ $= \sqrt{\frac{8}{3}G\pi r^2\rho} \quad (shown)$	
(b)	$v = \sqrt{\frac{8}{3}G\pi r^2\rho}$ $= \sqrt{\frac{8}{3}(6.67 \times 10^{-11})\pi(413 \times 10^3)^2(5500)}$ $= 724 \text{ ms}^{-1}$	
(c)	Assume that escape velocity v is equal to the root-mean-square speed c_{rms} of the argon molecules. By Kinetic Theory, average KE of one argon molecule = $\frac{3}{2} kT$ Average KE of one mole of monatomic gas is therefore $\frac{3}{2} RT$. $\frac{1}{2}M_r c_{rms}^2 = \frac{3}{2}RT_{\max} \quad \text{where is } M_r \text{ the molar mass of argon.}$ $T_{\max} = \frac{M_r c_{rms}^2}{3R}$ $= \frac{(40 \times 10^{-3})(724)^2}{3 \times 8.31}$ $= 841\text{K}$	
(d)	The root-mean-square speed of argon molecules will be less than escape velocity when temperature decreases. However, due to the random distribution of speeds, some molecules will have speeds greater than the escape velocity and would be able to escape.	

